MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 6 of 10 (Start time: 01:59:38 PM, 06 July 2025)

Integral of (1-2x) from [0,1] is .....

Select the correct option

○	1
◉	0
○	3
○	2

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 4 of 10 (Start time: 01:34:50 PM, 06 July 2025)

Arc length of the curve y=7 from x=0 to x=1 is _____.

Select the correct option

○	2
○	-1
◉	1
○	-2

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 5 of 10 (Start time: 01:44:30 PM, 06 July 2025)

If integral of 'f(x)' from [3,4] = - 8 ,then integral of 'f(x)' from [4,3] is .....

Select the correct option

○	9
◉	8
○	-6
○	-8

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 1 of 10 (Start time: 02:12:10 PM, 06 July 2025)

The Volume of a cylindrical shell is given by _____.

Select the correct option

- V = (average radius).(height).(thickness)
- V = 2π (average radius).(height).(thickness)

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 3 of 10 (Start time: 01:43:19 PM, 06 July 2025)

Arc length of the curve y=1 from x=a to x=b is .....

Select the correct option

○	0
○	a-b
◉	b-a
○	2(b-a)

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 6 of 10 (Start time: 01:44:43 PM, 06 July 2025)

Antiderivative of cosx is .....

Select the correct option

○	cosx+c
○	xsinx
○	cosx+sinx
●	sinx+c

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 1 of 10 (Start time: 01:42:47 PM, 06 July 2025)

Distance between (3,-2) and (4,0) using the distance formula is.....

Select the correct option

○	0
○	None of these.
◉	sqrt (5)
○	sqrt (3)

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 2 of 10 (Start time: 01:52:42 PM, 06 July 2025)

The volume of cylindrical shell for R = 5, r = 3 and h = 2 is ----------

Select the correct option

○	25*pi
○	30*pi
●	32*pi
○	36*pi

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 4 of 10 (Start time: 02:13:15 PM, 06 July 2025)

The volume V of a cylinder with base area A and height h is calculated by ---------

Select the correct option

○	V = A h^3
○	V = A h^2
○	V = 2A h
◉	V = A h

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 7 of 10 (Start time: 02:00:05 PM, 06 July 2025)

What is the geometric shape used in the method of cylindrical shells for finding volumes?

Select the correct option

◉	Cylinder.
○	Cone.
○	Pyramid.
○	Sphere.

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 1 of 10 (Start time: 01:51:58 PM, 06 July 2025)

What is a cylindrical shell?

Select the correct option

○	A sphere with a hole.
◉	A solid by two concentric cylinders.
○	A flat, two-dimensional shape.
○	A solid with a hole in it.

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 7 of 10 (Start time: 01:45:20 PM, 06 July 2025)

A plane curve (not line) is a curve that lies in a ..... plane.

Select the correct option

○	None of these
○	One dimensional
●	Two dimensional
○	Three dimensional

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 10 of 10 (Start time: 01:39:41 PM, 06 July 2025)

Find the area between $y = x$ *and* $y = -x(x-4)$.

Select the correct option

○	*None of these*
●	$\frac{9}{2}$
○	0
○	$\frac{7}{2}$

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 10 of 10 (Start time: 01:46:21 PM, 06 July 2025)

The area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Select the correct option

●	πab
○	$\frac{1}{4}\pi(a^2+b^2)$
○	$\pi(a+b)$
○	*None of these*

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 9 of 10 (Start time: 02:01:02 PM, 06 July 2025)

The volume of a cylindrical shell is given by ----------.

Select the correct option

○	$\left(\frac{\pi R^2}{r^2}\right)h$
○	$(\pi R^2 r^2)h$
○	$(\pi R^2 + \pi r^2)h$
◉	$(\pi R^2 - \pi r^2)h$

Question # 2 of 10 (Start time: 01:43:02 PM, 06 July 2025)

$$If\ f(x) = \sqrt{x},\ then \int_1^2 \pi [f(x)]^2\, dx\ is\ _____$$

Select the correct option

○	$\frac{7\pi}{2}$
○	$\frac{5\pi}{2}$
○	$\frac{\pi}{2}$
◉	$\frac{3\pi}{2}$

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 2 of 10 (Start time: 02:12:26 PM, 06 July 2025)

What is the purpose of using the washer method instead of the disk method?

Select the correct option

○	To increase accuracy
◉	To handle shapes with holes.
○	To simplify the integration process
○	To reduce the number of slices

Question # 8 of 10 (Start time: 02:00:19 PM, 06 July 2025)

Which of the following integrals represents the area between two curves $y = f(x) and\ y = g(x)$.

Select the correct option

○	$\int_a^b (f(x) + g(x))dx$
○	$\int_a^b \|f(x) + g(x)\|dx$
●	$\int_a^b (f(x) - g(x))dx$
○	$\int_a^b f(x)dx + \int_a^b g(x)dx$

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 5 of 10 (Start time: 02:13:39 PM, 06 July 2025)

Which of the following integrals represents the area between two curves $y = f(x) and\ y = g(x)$.

Select the correct option

◉	$\int_a^b (f(x) - g(x))dx$
○	$\int_a^b f(x)dx + \int_a^b g(x)dx$
○	$\int_a^b \lvert f(x) + g(x)\rvert dx$
○	$\int_a^b (f(x) + g(x))dx$

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 3 of 10 (Start time: 02:12:44 PM, 06 July 2025)

The volume by the washer perpendicular to the x-axis is

Select the correct option

○	$\int_a^b \pi([f(x)] + [g(x)])dx$
◉	$\int_a^b \pi([f(x)]^2 - [g(x)]^2)dx$
○	$\int_a^b \pi([f(x)]^2 + [g(x)]^2)dx$
○	$\int_a^b ([f(x)]^2 - [g(x)]^2)dy$

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 3 of 10 (Start time: 01:34:06 PM, 06 July 2025)

The area bounded by the parabola $y^2 = x$, *straight line* $y = 4$ *and* $y -$ *axis is*

Select the correct option

○	$7\sqrt{2}$
○	$\frac{16}{3}$
○	*None of these*
◉	$\frac{64}{3}$

Click to

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 10 of 10 (Start time: 02:01:34 PM, 06 July 2025)

Definite integral indicating the arc length of the curve y=coshx between x=0 and x=a is .....

Select the correct option

○	$\int_0^a \sinh x dx$
◉	$\int_0^a \cosh x dx$
○	None of these.
○	$\int_0^a 1+\sinh x dx$

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 2 of 10 (Start time: 01:32:37 PM, 06 July 2025)

Which integral represents the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ from x=2 to x=4 about y-axis by using cylindrical shells?

Select the correct option

○	$V = \int_2^4 2\pi x[\sqrt{x}]^2 dy$
◉	$V = \int_2^4 2\pi x\sqrt{x} dx$
○	$V = \int_2^4 2\pi x\sqrt{x} dy$
○	$V = \int_2^4 2\pi x[\sqrt{x}]^2 dx$

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 4 of 10 (Start time: 01:43:37 PM, 06 July 2025)

Definite integral indicating the arc length of the curve y=coshx between x=0 and x=a is .....

Select the correct option

◉	$\int_0^a \cosh x dx$
○	None of these.
○	$\int_0^a 1+\sinh x dx$
○	$\int_0^a \sinh x dx$

Question # 6 of 10 (Start time: 01:36:24 PM, 06 July 2025)

The volume by the washer perpendicular to the x-axis is

Select the correct option

○	$\int_a^b \pi([f(x)] + [g(x)])dx$
○	$\int_a^b \pi([f(x)]^2 + [g(x)]^2)dx$
◉	$\int_a^b \pi([f(x)]^2 - [g(x)]^2)dx$
○	$\int_a^b ([f(x)]^2 - [g(x)]^2)dy$

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 8 of 10 (Start time: 02:00:19 PM, 06 July 2025)

Which of the following integrals represents the area between two curves $y = f(x) and\ y = g(x)$.

Select the correct option

○	$\int_a^b (f(x) + g(x))dx$
○	$\int_a^b \|f(x) + g(x)\|dx$
○	$\int_a^b (f(x) - g(x))dx$
○	$\int_a^b f(x)dx + \int_a^b g(x)dx$

Question # 9 of 10 (Start time: 02:15:39 PM, 06 July 2025)

Which of the following integrals represents the surface area of the solid of revolution obtained by rotating the curve $y = f(x) = e^x$ around the x-axis over the interval [0,1]?

Select the correct option

○	$\int_0^1 2\pi e^x \sqrt{1+(e^x)^2}\,dx$
◉	$\int_0^1 2\pi x \sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$
○	$\int_0^1 2\pi e^x \sqrt{1+\frac{dy}{dx}}\,dx$
○	$\int_0^1 2\pi x \sqrt{1+\frac{dy}{dx}}\,dx$

Question # 4 of 10 (Start time: 01:58:06 PM, 06 July 2025)

If $w(y).v(y) \geq 0$ for all y *in* $[c, d]$, what can be concluded about the signs of $w(y)$ and $v(y)$?

Select the correct option

○	The signs of $w(y)$ *and* $v(y)$may vary for different y *in* $[c, d]$.
○	$w(y)$ *and* $v(y)$ have the same sign for all y *in* $[c, d]$.
○	$w(y)$ *and* $v(y)$ have opposite signs for all y *in* $[c, d]$.
○	$w(y)$ *and* $v(y)$ cannot have the same sign for all y *in* $[c, d]$.

Question # 4 of 10 (Start time: 01:58:06 PM, 06 July 2025)

If $w(y).v(y) \geq 0$ for all y *in* $[c, d]$, what can be concluded about the signs of $w(y)$ and $v(y)$?

Select the correct option

○	The signs of $w(y)$ *and* $v(y)$may vary for different y *in* $[c, d]$.
●	$w(y)$ *and* $v(y)$ have the same sign for all y *in* $[c, d]$.
○	$w(y)$ *and* $v(y)$ have opposite signs for all y *in* $[c, d]$.
○	$w(y)$ *and* $v(y)$ cannot have the same sign for all y *in* $[c, d]$.

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 5 of 10 (Start time: 01:59:06 PM, 06 July 2025)

The length of the WHOLE polygonal path will be _____.

Select the correct option

○	None of the above
◉	$\sum_{k=1}^{n} L_k = \sum_{k=1}^{n} \sqrt{1+(f'(x_k))^2}\Delta x_k$
○	$\sum_{k=1}^{n} L_k = \sum_{k=1}^{n} \sqrt{1+(f'(x_k))^2}$
○	$\sum_{k=1}^{n} L_k = \lim_{\max \Delta x \to 0} \sum_{k=1}^{n} \sqrt{1+(f'(x_k))^2}dx$

MTH101 - Calculus And Analytical Geometry (Quiz No. 5) Quiz Start Ti

Question # 5 of 10 (Start time: 01:35:19 PM, 06 July 2025)

What is the formula to calculate the surface area of a solid of revolution obtained by rotating a curve y=f(x) around the x-axis over the interval [a, b]?

Select the correct option

○	$\int_a^b 2\pi f(y)\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$
◉	$\int_a^b 2\pi f(x)\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$
○	$\int_a^b 2\pi f(x)\sqrt{1+\frac{dy}{dx}}dx$
○	$\int_a^b 2\pi y\sqrt{1+\frac{dy}{dx}}dx$

Click to Save Answe

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 8 of 10 (Start time: 01:45:36 PM, 06 July 2025)

Which of the following integrals represents the surface area of the solid of revolution obtained by rotating the curve $y = f(x) = e^x$ around the x-axis over the interval [0,1]?

Select the correct option

○	$\int_0^1 2\pi x \sqrt{1+\frac{dy}{dx}}dx$
◉	$\int_0^1 2\pi x \sqrt{1+\left(\frac{dy}{dx}\right)^2}dx$
○	$\int_0^1 2\pi e^x \sqrt{1+\frac{dy}{dx}}dx$
○	$\int_0^1 2\pi e^x \sqrt{1+(e^x)^2}\,dx$

MTH101 - Calculus And Analytical Geometry (Quiz No. 5) Quiz Start

Question # 7 of 10 (Start time: 01:37:14 PM, 06 July 2025)

Which statement accurately describes the comparison between the cylindrical shell method and the disk or washer method regarding their applications.

Select the correct option

○	The disk or washer method is simpler to apply.
◉	Both methods can be used interchangeably.
○	The cylindrical shell method is always more accurate.
○	Shell method is simpler to apply.

Question # 6 of 10 (Start time: 02:14:08 PM, 06 July 2025)

What is the formula to calculate the surface area of a solid of revolution obtained by rotating a curve y=f(x) around the x-axis over the interval [a, b]?

Select the correct option

Re

- $\int_a^b 2\pi f(x)\sqrt{1+\frac{dy}{dx}}dx$
- $\int_a^b 2\pi f(x)\sqrt{1+\left(\frac{dy}{dx}\right)^2}dx$
- $\int_a^b 2\pi y\sqrt{1+\frac{dy}{dx}}dx$
- $\int_a^b 2\pi f(y)\sqrt{1+\left(\frac{dx}{dy}\right)^2}dy$

Question # 9 of 10 (Start time: 01:38:47 PM, 06 July 2025) T

By using cylindrical shells to find the volume of the solid when the region R in the first quadrant enclosed between y=x and $y = x^2$ is revolved about the y-axis is _____.

Select the correct option

Reload Math

- $V = \int_0^1 2\pi(x - x^2)dx$
- $V = \int_0^1 2\pi x(x - x^2)dx$
- $V = \int_0^3 2\pi x(x - x^2)dx$
- $V = \int_0^1 \pi x(x - x^2)dx$

Click to Save Answer & Move to Nex

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Question # 9 of 10 (Start time: 01:46:07 PM, 06 July 2025)

If f(x) is a piecewise function with different expressions on different subintervals, how would you calculate the total area between the curves?

Select the correct option

○	Ignore the piecewise nature and integrate as a continuous function.
○	Take the average of f(x) over the interval [a, b].
◉	Apply the definite integral separately on each subinterval and sum the results.
○	Use the maximum value of f(x) within the interval.

Click to

MTH101 - Calculus And Analytical Geometry (Quiz No. 5) **Quiz Start Time: 01:32 PM, 06 July 2025**

Question # 8 of 10 (Start time: 01:37:30 PM, 06 July 2025) **Total Marks:** 1

The volume of the solid generated by the region enclosed between $y = \sqrt{x}$ x=1 , x=4 and x-axis is revolved about y-axis.Which of the following equation gives the volume of solid by cylindrical shell _____.

Select the correct option

Reload Math Equations

○	$\frac{114\pi}{5}$
◉	$\frac{124\pi}{5}$
○	77.91
○	Both a and c

MTH101 - Calculus And Analytical Geometry (Quiz No. 5) Quiz Start Time: 02:12 PM, 06 July 2025

Question # 10 of 10 (Start time: 02:16:06 PM, 06 July 2025) **Total Marks:** 1

The volume of the solid generated by the region enclosed between $y = \sqrt{x}$ x=1 , x=4 and x-axis is revolved about y-axis.Which of the following equation gives the volume of solid by cylindrical shell ____.

Select the correct option

Reload Math Equations

- $\frac{114\pi}{5}$
- $\frac{124\pi}{5}$
- 77.91
- Both a and c

Click to Save Answer & Move to Next Question

sec(s)

MTH101 - Calculus And Analytical Geometry (Quiz No. 5) Quiz Start Time: 01:32 PM, 06 July 2025

Question # 2 of 10 (Start time: 01:32:37 PM, 06 July 2025) **Total Marks:** 1

Which integral represents the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ from x=2 to x=4 about y-axis by using cylindrical shells?

Select the correct option

Reload Math Equations

- $V = \int_2^4 2\pi x[\sqrt{x}]^2 dy$
- $V = \int_2^4 2\pi x\sqrt{x}dx$
- $V = \int_2^4 2\pi x\sqrt{x}dy$
- $V = \int_2^4 2\pi x[\sqrt{x}]^2 dx$

Click to Save Answer & Move to Next Question

MTH101 - Calculus And Analytical Geometry (Quiz No. 5) Quiz Start Time: 02:12 PM, 06 July 2025

Question # 7 of 10 (Start time: 02:14:26 PM, 06 July 2025) **Total Marks:** 1

The volume of the solid generated by the region enclosed between $y = \sqrt{x}$ x=1 , x=3 and x-axis is resvolved by y-axis.Which of the following equation gives the volume of solid by cylindrical shell ____.

Select the correct option

Reload Math Equations

- Both b and c
- $V = \int_1^3 2\pi x\sqrt{x}\,dx$
- $V = \int_1^3 2\pi x^{\frac{3}{2}}\,dx$
- $V = \int_1^x 2\pi x\sqrt{x}\,dx$

Click to Save Answer & Move to Next Question

sec(s)

MTH101 - Calculus And Analytical Geometry (Quiz No. 5)

Quiz Start Time: 02:12 PM, 06 July 2025

Question # 8 of 10 (Start time: 02:15:06 PM, 06 July 2025)

Total Marks: 1

$Find\ the\ area\ of\ the\ region\ to\ the\ left\ of\ the\ parabola\ x = 2y^2,\ to\ the\ right\ of\ the\ y-axis\ and\ between\ y = 1\ and\ y = 3$

Select the correct option

Reload Math Equations

○	$\frac{1}{3}$
○	$\frac{10}{4}$
◉	$\frac{52}{3}$
○	$None\ of\ these$

Click to Save Answer & Move to Next Question